Hassan R. S. Andrabi

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EDUCATION

PhD	Decision, Risk, and Financial Sciences, University of Melbourne, 2027
MCom	Decision, Risk, and Financial Sciences, University of Melbourne, 2024
Hons	Finance, University of Melbourne, 2020
BCom	Commerce, University of Melbourne, 2019

RESEARCH AREAS

Multi-agent AI: strategic interaction, coordination, and evaluation of LLM-agent systems.

Markets: collective intelligence, the role of incentives, decentralised search and discovery.

Computational complexity: combinatorial problems, phase transitions, and human/machine problem-solving.

WORK IN PROGRESS

H. Andrabi, P. Bossaerts, C. Murawski, N. Yadav. "Aigora: Experimental Markets with LLM agents."

H. Andrabi, P. Bossaerts, N. Yadav. "Can Markets Coordinate Discovery?"

H. Andrabi, P. Bossaerts, N. Yadav. "Cognitive Foundations of Choice under Computational Complexity."

H. Andrabi, P. Bossaerts, K. Ioannidis, R. Woods, N. Yadav. "Economic Oracles: How Markets Reveal Optimal Solutions to Complex Problems."

PUBLICATIONS

Journal Articles

2025 S. Bode, A. Varsamis, Z. A. Xu, M. Jiwa, H. Andrabi, N. Egorova-Brumley, C. Morawetz. "When We Want to Know the Bad News: Exploring Information-Seeking for Unavoidable Pain Stimuli." *Cognition and Emotion*. doi.org/10.1080/02699931.2025.2597887

2024 D. Ait Ouakrim, T. Wilson, A. Waa, R. Maddox, H. Andrabi, S. R. Mishra, J. A. Summers, C. E. Gartner, R. Lovett, R. Edwards, N. Wilson, T. Blakely. "Tobacco Endgame Intervention Impacts on Health Gains and Maori: Non-Maori Health Inequity: A Simulation Study of the Aotearoa/New Zealand Tobacco Action Plan." *Tobacco Control* 2 (33), 173-184. doi.org/10.1136/tc-2022-057655

2023 S. R. Mishra, T. Wilson, H. Andrabi, D. Ait Ouakrim, A. Li, E. Akpan, R. Bentley, T. Blakely. "The Total Health Gains and Cost Savings of Eradicating Cold Housing in Australia." *Social Science & Medicine* 334 (115954). doi.org/10.1016/j.socscimed.2023.115954

2023 J. Szanyi, T. Wilson, S. Howe, J. Zeng, H. Andrabi, S. Rossiter, T. Blakely. "Epidemiologic and Economic Modelling of Optimal COVID-19 Policy: Public Health and Social Measures, Masks and Vaccines in Victoria, Australia." *The Lancet Regional Health – Western Pacific* 32 (100675). doi.org/10.1016/j.lanwpc.2022.100675

INVITED TALKS

Seminars and Workshops

2026 "Incentives for Discovery in Computationally Hard Environments." AFAANZ Doctoral Symposium. Accounting and Finance Association of Australia and New Zealand (AFAANZ). Melbourne, Australia. Jul 5.

2026 "Incentives for Discovery in Computationally Hard Environments." Workshop on Computational Complexity and Economic Decision Making. University of Cambridge. Cambridge, UK. May 28.

2026 "The Impact of Price Signals on Effort and Productivity in a 'Hard' Task." Workshop on Dynamic Decision Making: Minds, Models, and Markets. The University of Sydney. Sydney, Australia. Apr 14.

2025 "Phase Transition in the 0-1 Knapsack Problem." Leverhulme International Professorship in Neuroeconomics Complexity Seminar. University of Cambridge. Cambridge, UK. Mar 3.

2024 "Prices as Information Signals." Leverhulme International Professorship in Neuroeconomics Complexity Seminar. University of Cambridge. Cambridge, UK. Nov 19.

Conference Activity[1]

2026 H. Andrabi, P. Bossaerts, N. Yadav. "Incentives for Discovery in Computationally Hard Environments." Society for Experimental Finance. Toulouse, France. Jun 15.

2026 H. Andrabi, P. Bossaerts, N. Yadav. "Incentives for Discovery in Computationally Hard Environments." International Association for Research in Economic Psychology, Motives and Preferences in Economic Decisions. Castellón, Spain. Jun 9.

2026 H. Andrabi, *P. Bossaerts*, N. Yadav. "Cognitive Foundations of Choice under Computational Complexity." International Association for Research in Economic Psychology, Motives and Preferences in Economic Decisions. Castellón, Spain. Jun 9.

2026 H. Andrabi, P. Bossaerts, N. Yadav. "The Effect of Price Signals on Effort and Productivity in a 'Hard' task." Asia-Pacific Meeting of the Economic Science Association. Melbourne, Australia. Apr 9.

2026 H. Andrabi, P. Bossaerts, N. Yadav. "The Invisible Hand: How Markets Guide Effort and Productivity" Experimental Finance Asia/Pacific. Melbourne, Australia. Apr 8.

2026 H. Andrabi, P. Bossaerts, N. Yadav. "The Impact of Price Signals on Effort and Productivity in a 'Hard' Task." Australasian Mathematical Psychology Conference. Singapore. Feb 27.

2025 H. Andrabi, P. Bossaerts, *K. Ioannidis*, R. Woods, N. Yadav. "Markets or Patents? Experiments on Collective Innovation and Information Aggregation" Society for Experimental Finance Conference. Maastricht, Netherlands. Jun 12.

2024 H. Andrabi, P. Bossaerts, *K. Ioannidis*, R. Woods, N. Yadav. "Markets spread knowledge to overcome computational complexity." Society for Experimental Finance Conference. Stavanger, Norway. Jun 25.

2024 H. Andrabi, P. Bossaerts, K. Ioannidis, R. Woods, N. Yadav. "Markets Reveal Optimal Solutions to Complex Problems." Society for Neuroeconomics Conference. Cascais, Portugal. Oct 11.

GRANTS AND AWARDS

Awards and Honors

2026 Research Travel Scholarship (M. A. Bartlett) for High Achieving Students, Faculty of Business and Economics, University of Melbourne. $2,200 AUD.

2026 Health Promotion Grant, awarded for a Faculty level wellbeing initiative., University of Melbourne. $9,000 AUD.

[1]Presenting author *italicised* if other than first author.

2025 Vernon Smith Young Talent Award, Society for Experimental Finance. $4,000 USD.

2024 Melbourne Global Scholars Award for High Achieving Students, Faculty of Business and Economics, University of Melbourne. $1,500 AUD.

2023 Best Research Paper by Discipline Award (Methodology), School of Population and Global Health, University of Melbourne. $1,500 AUD.

2022 MPSGH Research Achievement Award, School of Population and Global Health, University of Melbourne. $1,500 AUD.

Competitive Scholarships

2026 Graduate Research Travelling Scholarship, Faculty of Business and Economics, University of Melbourne. $4,000 AUD.

2024 International Study Scholarship, Faculty of Business and Economics, University of Melbourne. $5,000 AUD.

2023 Doctoral Program Scholarship, Faculty of Business and Economics, University of Melbourne. $37,000 AUD p.a.

2014 Kwong Lee Dow Scholarship for High Achieving Students, University of Melbourne. $2,500 AUD.

TEACHING

2026 Foundations of Business Analytics (undergraduate), TA; Behavioural Finance (undergraduate), TA; Principles of Finance (undergraduate), TA

2025 Machine Learning in Finance (postgraduate), head TA; Principles of Finance (undergraduate), TA

2024 Machine Learning in Finance (postgraduate), head TA; Algorithmic Trading (undergraduate), TA; Capstone Studies in Finance (postgraduate), TA

2023 Machine Learning in Finance (postgraduate), TA

2021 Contemporary Epidemiology Short Course (postgraduate), Instructor

PROFESSIONAL EMPLOYMENT

2025– Software Engineer (frontend), Centre for Brain, Mind, & Markets, University of Melbourne

2020–23 Software Engineer (Simulation Modelling), Population Interventions Unit, University of Melbourne

2019–20 Accounting and Advisory, MJC Partners

SERVICE

Service to the Field

2025 PyKP, a Python package for the 0-1 Knapsack Problem. `https://pykp.readthedocs.io/`.

2023–24 Managing Editor and Software Engineer, Australian COVID-19 Modelling Initiative (AUSCMI)

2022 CovStrain, a web-based tool to track global genomic sequencing data on SARS-CoV-2 mutations of interest. 1k users. `https://covstrain.vercel.app/`.

2020 COVIDVictoria, a web-based tool for COVID-19 Spatial Data Mapping. 2m users.

Service to the University

2026 Secretary, Faculty of Business and Economics PhD Student Society, University of Melbourne

SKILLS

Programming: Python, R, Julia, React, JavaScript, Git, high-performance computing

Languages: English (native), Urdu (fluent)

Updated November 2026